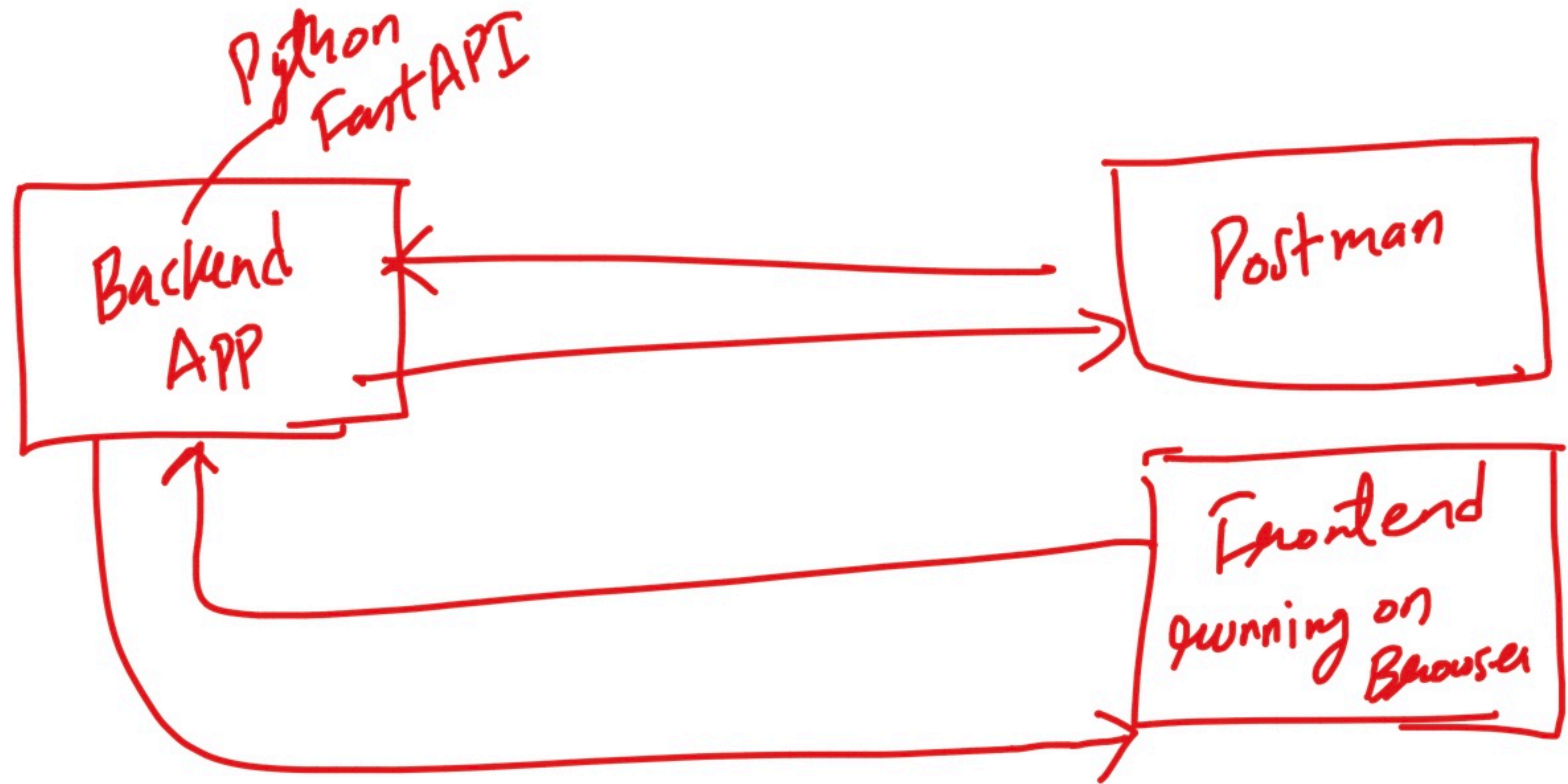
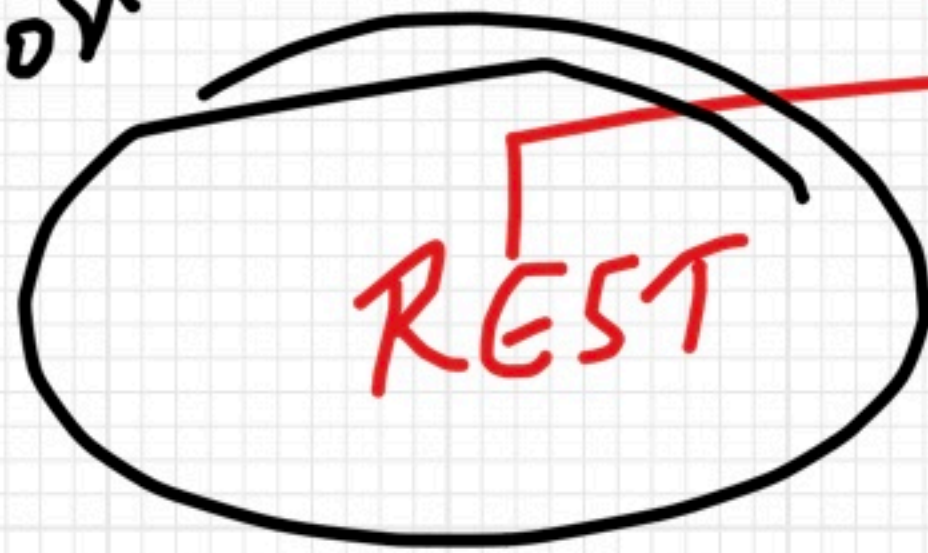




Most widely used



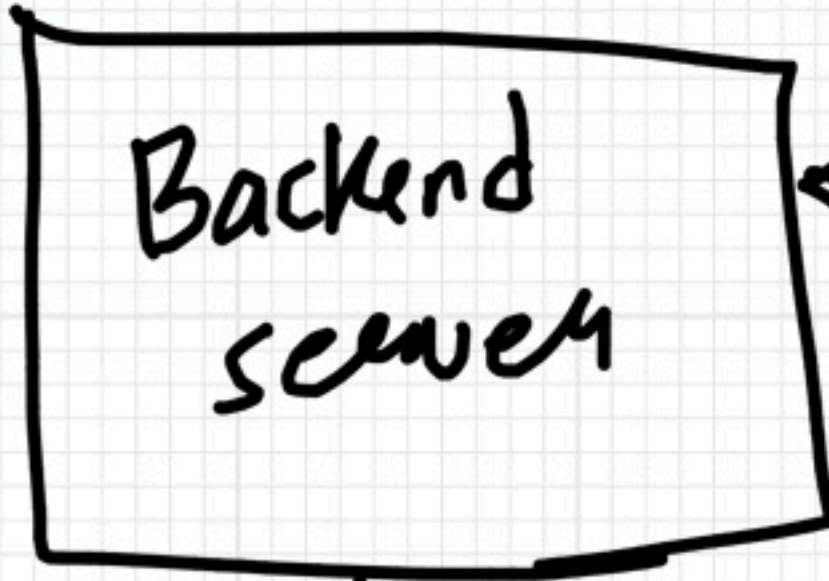
APIs

SOAP

gRPC

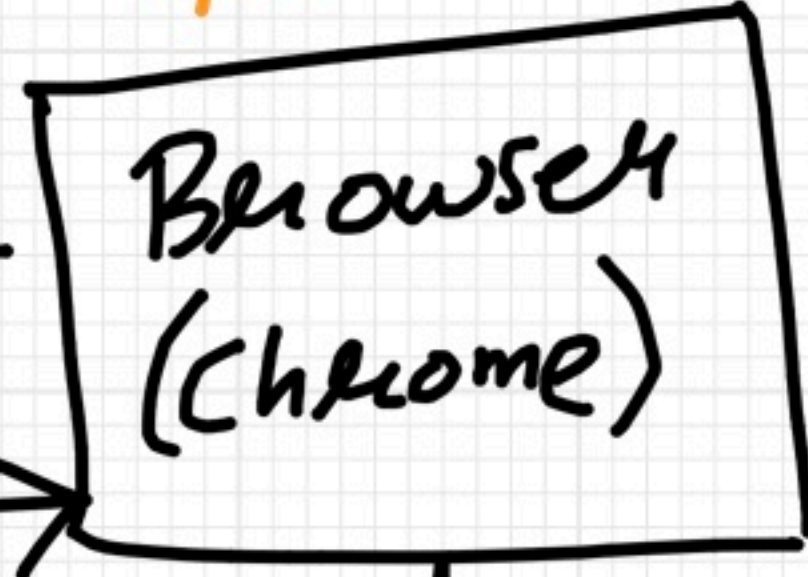
GraphQL, etc

Python
Fast API



get/products

JavaScript
React



[{"prod_id": 1},
{"prod_id": 2}]

FastAPI

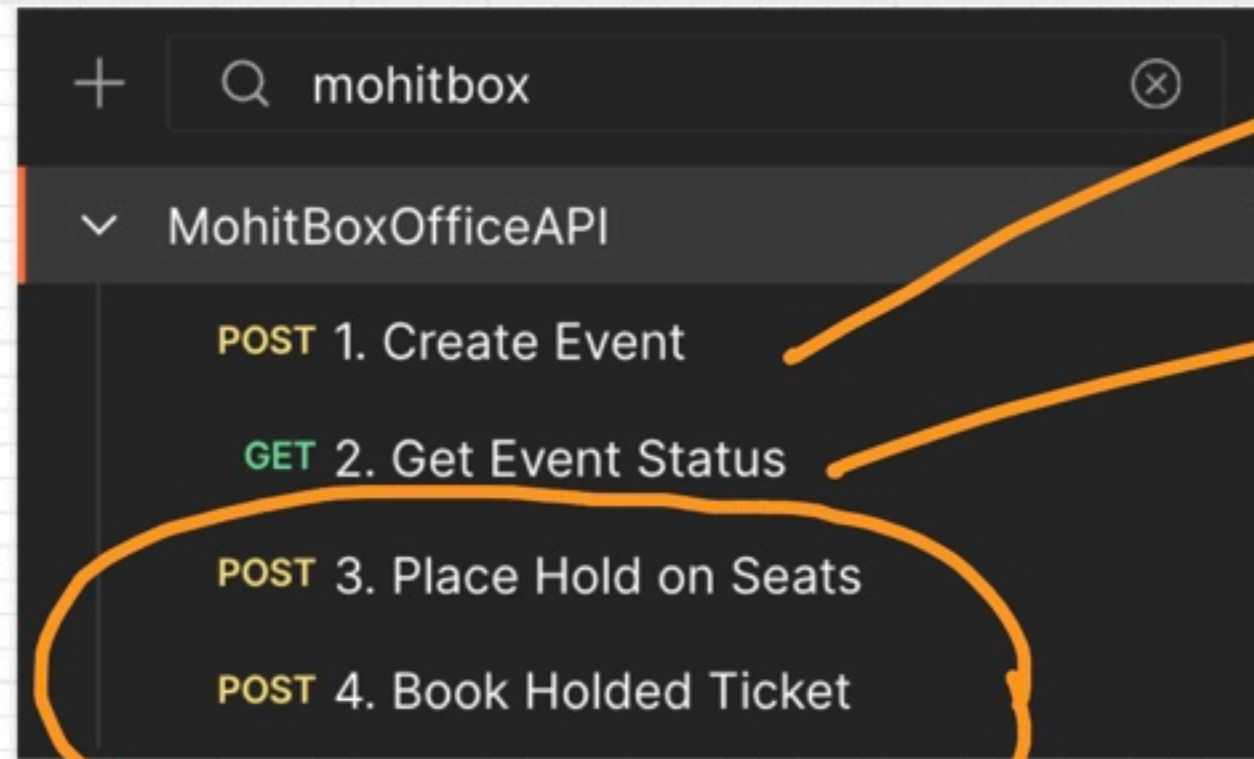
→ Web framework

Getting popularity

	django	 Flask	 FastAPI
Performance speed	Normal	Faster than Django	The fastest out there ✓
Async support	YES with restricted latency	NO needs Asyncio	YES native async support ✓
Packages	Plenty for robust web apps	Less than Django for minimalistic apps	The least of all for building web apps faster ✓
Popularity	The most popular	The second popular	Relatively new ✓
Learning	Hard to learn	Easier to learn	The easiest to learn ✓

Sample Fast API Project: MohitBoxOFFICE

Create Event like Python Session by Mohit



Get status of how many seats are booked, etc

Customer APIs to hold seats, and then book them

Admin API's

Molit Box Office Requirements

Core APIs

①

- **POST /events**

Create an event: { **name**, **total_seats** } → returns { **event_id**, **total_seats**, **created_at** }.

②

- **POST /holds**

Request a temporary hold on seats for **2 minutes**: { **event_id**, **qty** } → returns { **hold_id**, **expires_at**, **payment_token** }.

- The hold must decrement *available* seats immediately.
- If there aren't enough seats, return a clear 4xx error.
- Multiple concurrent hold requests must not oversubscribe seats.

③

- **POST /book**

Confirm a booking using an active hold: { **hold_id**, **payment_token** } → returns { **booking_id** }.

- **Idempotent**: If the same request is retried (e.g., network glitch), it must not double-book.

④

- **GET /events/{id}**

Returns snapshot: { **total**, **available**, **held**, **booked** }.

Behavior & Constraints

- **Auto-expiry**: Holds expire after **2 minutes** and release seats automatically.
- **No overbooking**: Under parallel requests, total booked + active holds must never exceed total seats.
- **Storage**: In-memory or lightweight DB is fine (document your choice).

Testing & Debugging

1. The code to be tested

Create a Python file, for example, `my_math.py`, with the function you want to test. 

python

```
# my_math.py
def add_numbers(a, b):
    return a + b
```

Use code with caution.



2. The test file

Create a test file, such as `test_my_math.py`. Your test class must inherit from `unittest.TestCase`, and test methods must start with `test_`. 

python

```
# test_my_math.py
import unittest
from my_math import add_numbers

class TestAddNumbers(unittest.TestCase):
    def test_add_positive_numbers(self):
        self.assertEqual(add_numbers(2, 3), 5)

    def test_add_negative_numbers(self):
        self.assertEqual(add_numbers(-2, -3), -5)

    def test_add_zero(self):
        self.assertEqual(add_numbers(0, 5), 5)

if __name__ == '__main__':
    unittest.main(verbosity=2)
```

sh

```
python -m unittest test_my_math.py
```